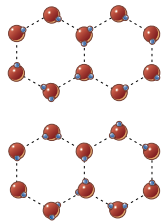




▲ FLOATING ICE

As water cools it gets more dense and heavy, so it sinks. However, when it turns to ice, it grows less dense and floats, because its molecules spread to form a rigid hexagonal lattice, like honeycomb. In the process they expel salt, so the water in sea ice is almost fresh.



MOLECULES FREEZE INTO CRYSTAL LATTICE

ICE FORMATION AT SEA

As the polar winter approaches, air temperatures fall and make water at the sea surface freeze into small crystals of frazil ice. If these are not dispersed by waves they gradually congeal into soupy grease ice.

As temperatures keep falling, the ice forms a thin layer at the surface. Water movement makes this break up into small rafts that rub together and develop the raised edges of pancake ice.

Eventually, the pancake ice freezes into a solid sheet, which gets steadily thicker all winter. During the following summer the sheet breaks up into large ice floes that drift with the winds and currents as pack ice. These floes often get pushed together to form a continuous, tumbled mass of floating ice.

GREASE ICE



PANCAKE ICE



PACK ICE



ICY OCEANS

In the polar regions, the sea freezes where it is in contact with the extremely cold air. The ice floats at the surface instead of sinking to the ocean floor, because although it is colder than water, it is less dense. It can cover vast areas of ocean, creating the shifting, frozen world of pack ice. The ice cover expands in the dark, bitterly cold polar winter, and this affects the temperature, saltiness, and density of the ocean water below. But a lot of the ice melts away in summer when there is almost constant daylight, allowing light to flood the polar oceans and trigger an explosion of marine life.



► ICEBREAKERS

In the Arctic Ocean, massive reinforced icebreakers plow through the sea ice to maintain corridors of open water for shipping. They ride up over the ice and smash it with their immense weight. They cannot deal with really dense pack ice, however, so they have to keep breaking the ice to stop it from growing too thick.



▲ DRIFTING FLOES

In some places like the central Arctic Ocean, pack ice form dense masses that are often piled into pressure ridges. Yet even these apparently solid sheets of ice keep moving. When Ernest Shackleton's (1874–1922) *Endurance* was trapped in the thick pack ice of the Antarctic Weddell Sea in 1915, the ship drifted 800 miles (1,300 km) with the ice before it was crushed and destroyed.